INCH-POUND
MIL-DTL-16377/74A
30 June 2014
SUPERSEDING
MIL-DTL-16377/74(SH)
27 September 2004

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, EMERGENCY LIGHTING, 40 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 77.4EM, 77.5EM, 344.2EM, 344.3EM, 77.4EMSSL, 77.5EMSSL, 344.2EMSSL, AND 344.3EMSSL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

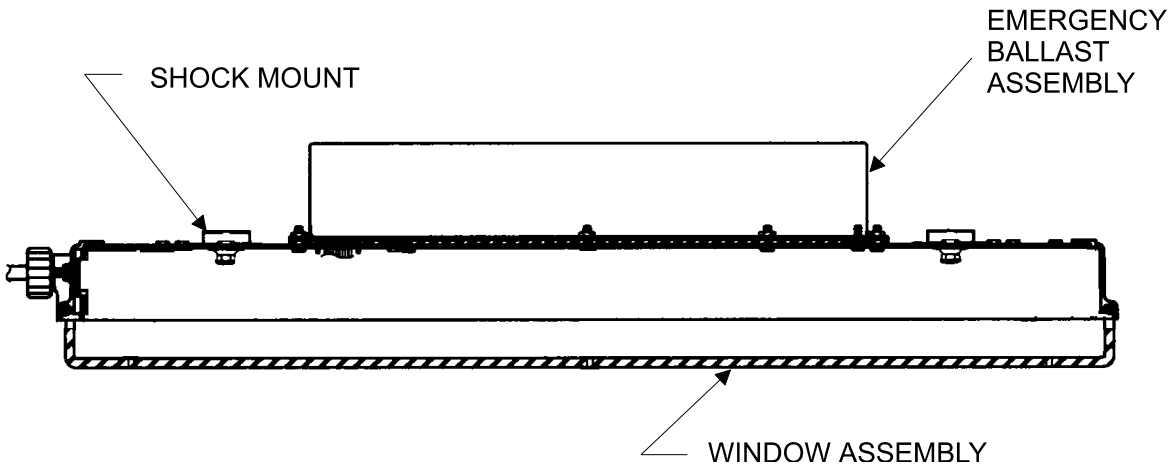
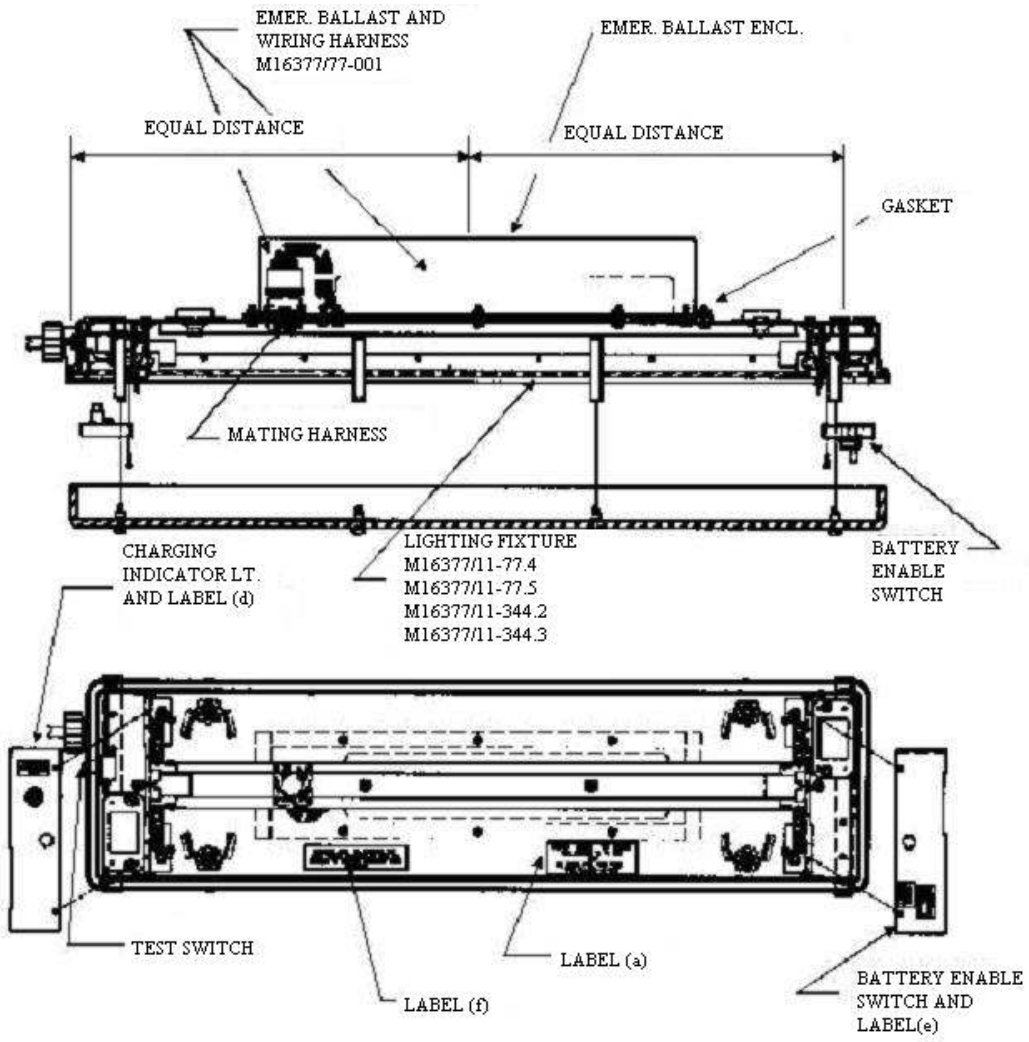


FIGURE 1. Typical outline of symbols 77.4EM, 77.5EM, 344.2EM, and 344.3EM.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

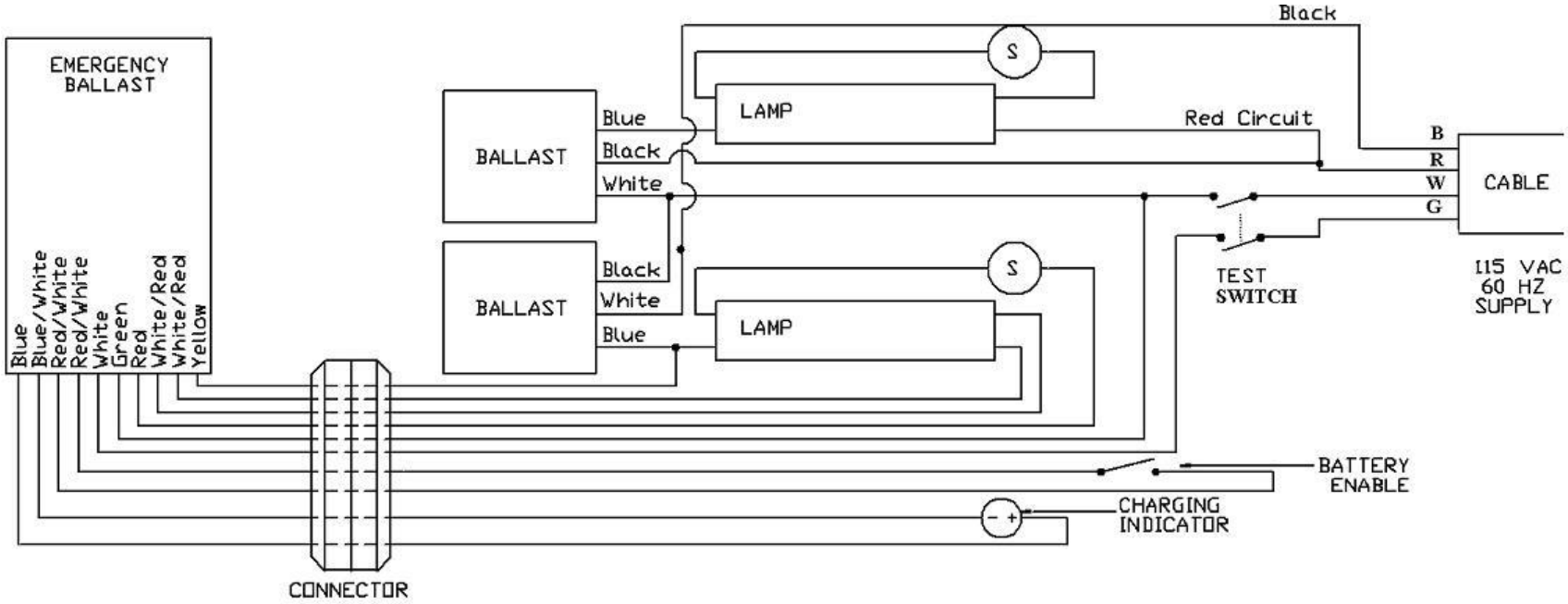


NOTES:

1. For dimensional data, see figures 1, 2, and 3 of MIL-DTL-16377/11.
2. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

FIGURE 2. Typical view of symbols 77.4EM, 77.5EM, 344.2EM, and 344.3EM.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.



NOTES:

1. Ballast black and white wiring not color coded with input cable black and white, emergency lamp only.
2. Emergency ballast white and green wiring not color coded with input cable white and green.

FIGURE 3. Wiring diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

77.4EM: Complete lighting fixture assembly, standard mounting, with two 20-watt cool white fluorescent lamps, white diffusing window assembly, and fluorescent emergency ballast.

77.5EM: Complete lighting fixture assembly, flush mounting, with two 20-watt cool white fluorescent lamps, white diffusing window assembly, and fluorescent emergency ballast.

344.2EM: Complete lighting fixture assembly, standard mounting, with two 20-watt cool white fluorescent lamps, clear and white prismatic window assembly, and fluorescent emergency ballast.

344.3EM: Complete lighting fixture assembly, flush mounting, with two 20-watt cool white fluorescent lamps, clear and white prismatic window assembly, and fluorescent emergency ballast.

DESIGN:

Electrical rating:	40 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, and 3 of MIL-DTL-16377/11 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 4 of MIL-DTL-16377/11.
Lampholder spacing:	See figure 21 of MIL-DTL-16377.
Wiring diagram:	See figure 3.
Mounting dimensions:	
Symbols 77.4EM and 344.2EM:	See figure 25 of MIL-DTL-16377.
Symbols 77.5EM and 344.3EM:	See figure 26 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Gasket:	Neoprene rubber (durometer 35-45).
Weight:	
Symbols 77.4EM and 344.2EM:	17 pounds maximum.
Symbols 77.5EM and 344.3EM:	19 pounds maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/11. Labels (d), (e), and (f) as shown on figure 2:

a. Two Type I labels.
b. One Type II label for Symbols 77.5EM and 344.3EM only.
c. One Type IV label. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I, as by name and stock number only.
d. One Type IV label. The information to be shown on the Type IV label shall be "CHARGING INDICATOR LIGHT."
e. One Type IV label. The information to be shown on the Type IV label shall be "BATTERY ENABLE."
f. One Type V label. The information to be shown on the Type V label shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

Standard parts: The standard parts and their respective part numbers shown in table I shall be used in the construction of the fixtures specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

77.4EMSSL:	Complete lighting fixture assembly, standard mounting, with LED light source, white diffusing window assembly, and fluorescent emergency battery back-up.
77.5EMSSL:	Complete lighting fixture assembly, flush mounting, with LED light source, white diffusing window assembly, and emergency battery back-up.
344.2EMSSL:	Complete lighting fixture assembly, standard mounting, with LED light source, clear and white prismatic window assembly, and emergency battery back-up.
344.3EMSSL:	Complete lighting fixture assembly, flush mounting, with LED light source, clear and white prismatic window assembly, and emergency battery back-up.

DESIGN:

Electrical rating:	40 watts maximum, 120 volts, 60 hertz.
Dimension and configurations:	External dimensions shall not exceed those shown on figures 1, 2, and 3 of MIL-DTL-16377/11 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 4 and table I of MIL-DTL-16377/11 and 3.7.1 of MIL-DTL-16377.
Emergency mode lumen output:	Luminaire shall provide a minimum of 240 lumens on a fully charged battery and maintain over 144 lumens over a 5-hour runtime.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Shall maintain same control scheme as legacy fluorescent, including battery enable switch, battery charging indicator.
Mounting dimensions:	
Symbols 77.4EMSSL and 344.2EMSSL:	See figure 25 of MIL-DTL-16377.
Symbols 77.5EMSSL and 344.3EMSSL:	See figure 26 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Gasket:	Neoprene rubber (durometer 35-45), if design uses legacy fluorescent emergency ballast enclosure.
Weight:	
Symbols 77.4EMSSL and 344.2EMSSL:	17 pounds maximum.
Symbols 77.5EMSSL and 344.3EMSSL:	19 pounds maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377. Labels (a), (b), and (c) in accordance with MIL-DTL-16377/11. Labels (d), (e), and (f) as shown on figure 2:

a. Two Type I labels.
b. One Type II label for Symbols 77.5EM and 344.3EM only.
c. One Type IV label. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by:
   (1) A list of all standard parts shown in table I by name and stock number only, if appropriate, or

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

(2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

d. One Type IV label. The information to be shown on the Type IV label shall be "CHARGING INDICATOR LIGHT."

e. One Type IV label. The information to be shown on the Type IV label shall be "BATTERY ENABLE."

f. One Type V label. The information to be shown on the Type V label shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

BATTERY BACK-UP SYSTEM REQUIREMENTS:

Battery back-up system for Type III luminaires.

DESIGN:

Assembly:	The battery back-up system consists of a high temperature, rechargeable battery; battery charger; AC power sensing circuit; wiring harness; enclosure.
Circuit design:	Back-up system shall incorporate Thyristor devices for protection against high voltage spikes.
Battery back-up location:	Battery back-up assembly shall fit within the dimensions of figure 1 of MIL-DTL-16377/8 or within the legacy fluorescent emergency ballast enclosure, as described in table I.
Battery material:	Use of other than NI-CAD battery systems requires specific authorization from NAVSEA.
Emergency operation time:	5 hours minimum.
Full recharge time:	20 hours maximum.
Temperature rating:	131 °F (55 °C) minimum.
Charging power:	4.0 watts maximum.
System power factor:	>0.95.
System harmonics:	<3 percent.
Wiring:	Connection between battery back-up system and the rest of the luminaire shall be made with a commercial wiring harness that allows for easy replacement of battery back-up system.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for Type I fixtures and table III for Type III luminaires.
Conformance:	
Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

TABLE I. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Complete fixture, Symbol 77.4	M16377/11-77.4	6210-01-465-6684	For use with Symbol 77.4EM
Complete fixture, Symbol 77.5	M16377/11-77.5	6210-01-088-2510	For use with Symbol 77.5EM
Complete fixture, Symbol 344.2	M16377/11-344.2	6210-01-149-4168	For use with Symbol 344.2EM
Complete fixture, Symbol 344.3	M16377/11-344.3	6210-01-088-6185	For use with Symbol 344.3EM
Window, white diffusing	M16377/11-001	Window assembly, white diffusing	6210-00-399-7035
Window, clear and white prismatic	M16377/11-003	Window assembly, clear and white prismatic	6210-01-169-5997
Mating harness	AMP part No. 206043-1 or equal		Mates with M16377/77-001
Emergency ballast and wiring harness	M16377/77-001	6250-01-491-7039	
Indicator light	Commercial	6220-01-415-7889	LED type
Battery enable switch [1/]	Commercial	5930-01-414-9240	DPST
Enclosure for emergency ballast			Aluminum; approximately 3″ H x 3 ½″ W x 16″ L
Test switch, push	Commercial	5930-01-416-0342	Double pole, test switch with rubber boot
Switch cover, rubber boot	MIL-DTL-5423/2		Neoprene
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-005	6210-00-441-5138	Not furnished, see 3.7.9 of MIL-DTL-16377
Amber filter tube	M16377/42-006	6210-01-166-7184	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-007	6210-01-498-8001	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-009	6210-01-313-8168	Not furnished, see 3.7.9 of MIL-DTL-16377

NOTE:

[1/] Switch shall normally be left in the “Battery Enable” position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

TABLE II. First article and comparison inspection for complete type I fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	Candlepower
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

TABLE III. First article and comparison inspection for complete type III luminaires. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.14	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output	3.7.1.1	4.8.6.8	Lumen output
Emergency battery back-up performance: Operating time and discharge test	3.6.19.3	4.8.26	

NOTES:

1/ Requirements and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.

TABLE IV. Part number and general information.

Military Part Number M16377/74-	Item Name	National Stock Number (NSN)
77.4EM	Complete fixture, Symbol 77.4EM	6210-01-508-6411
77.5EM	Complete fixture, Symbol 77.5EM	6210-01-568-8795
344.2EM	Complete fixture, Symbol 344.2EM	
344.3EM	Complete fixture, Symbol 344.3EM	
77.4EMSSL	Complete fixture, solid state, Symbol 77.4EMSSL	
77.5EMSSL	Complete fixture, solid state, Symbol 77.5EMSSL	
344.2EMSSL	Complete fixture, solid state, Symbol 344.2EMSSL	
344.3EMSSL	Complete fixture, solid state, Symbol 344.3EMSSL	

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-034)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:54Z
Check the source to verify that this is the current version before use.